

Problem Set 5

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1 Questions and Responses

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Table 1: Comparison of Estimated Coefficients Across Methods

	True β	Closed-Form OLS	Gradient Descent	L-BFGS	Nelder-Mead	MLE
β_1	1.500	1.501	1.501	1.501	1.501	1.501
β_2	-1.000	-1.001	-1.001	-1.001	-1.001	-1.001
β_3	-0.250	-0.252	-0.252	-0.252	-0.252	-0.252
β_4	0.750	0.749	0.749	0.749	0.749	0.749
β_5	3.500	3.501	3.501	3.501	3.501	3.501
β_6	-2.000	-2.001	-2.001	-2.001	-2.001	-2.001
β_7	0.500	0.499	0.499	0.499	0.499	0.499
β_8	1.000	1.003	1.003	1.003	1.003	1.003
β_9	1.250	1.247	1.247	1.247	1.247	1.247
β_{10}	2.000	2.001	2.001	2.001	2.001	2.001
σ	0.500					0.500

Notes: The table reports the true parameter values and the estimated coefficients from the closed-form OLS solution, gradient descent, `nloptr` with L-BFGS, `nloptr` with Nelder-Mead, and maximum likelihood estimation. The MLE estimate of σ is reported in the final row.